



2026 Hardware Tutorial Syllabus

After the Welcoming Session, weekly tutorials will be held over four weeks to help trainees learn and become familiar with robotics electronics, schematic design, and PCB design. Attendance will be taken for each session.

Week 1: Introduction to Hardware & Schematic Design

- Introduction to hardware structure and the role of hardware team members
- Robot hardware structure (ie. battery, relay, emergency stop, wiring, motors, sensors)
- Circuit-board structure breakdown
- Passive components and their typical applications (ie. resistors, capacitors, diodes)
- Board components by function (ie. power, I/O, MOSFETs, motor control)
- Basic datasheet reading
- Introduction to schematic design with KiCad
- Introduction to MCUs

Week 2: Introduction to PCB Design

- Converting a schematic into a PCB using KiCad
- Basic KiCad PCB controls and workflow
- Introduction to PCB footprints and board layout
- Demonstration of component-placement considerations
- Introduction to the PCB project

Week 3: Soldering Techniques

- Laboratory equipment and safety procedures
- Hands-on soldering practice with debugging
- Through-hole components (ie. resistors, capacitors, LEDs)
- SMD components (ie. 0603 components)
- Multi-pin components (ie. MCUs, USB Type-C connectors)
- Basic inspection, rework, DMM use, and continuity testing

Week 4: Wire Making

- Creating power cables (ie. XT60 to XT30)
- Signal-wire creation
- Wire crimping (ie. XH connectors)
- Connecting wires
- Basic debugging and connection testing